

OWCA

Organization Without a Cool Attitude *Database Design & Implementation Report*

CS 470(G) Database Systems

Spring 2026 Final Project

TEAM_2

Josh Roth

Rishabh Arya

Oluwanifemi Yesufu

1. Organization Overview

OWCA (Organization Without a Cool Attitude) is an agency directly inspired by the animated series Phineas and Ferb. In that universe, OWCA is a secret government organization that recruits animal agents to thwart the schemes of supervillains known as "nemeses." Similarly, in our scenario, agents live undercover as ordinary household pets within unsuspecting families, reporting to human handlers through a network of hidden field offices. Nemeses build doomsday devices called "inators" in secret lairs, which agents must neutralize before the day is out. This database models the lore behind that world.

OWCA is organized around three main roles:

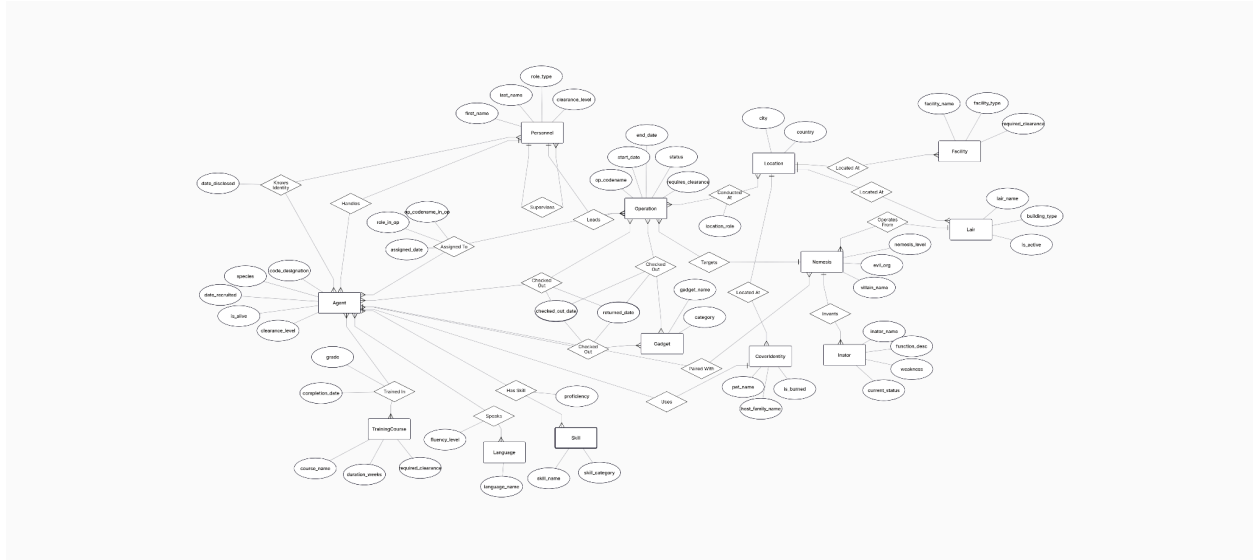
1. Director: Oversees operations at the classification level, reviews nemesis threat profiles, and allocates resources.
2. Handler: Operational coordinator. Assembles agent rosters, assigns missions, manages gadget logistics, and manages safe-house access.
3. Analyst: Intelligence specialist. Traces identity leaks, maps inter-agent relationships, assesses damage from compromised operations, and keeps tracks of the nemeses.

More introductory information can be found here:

https://phineasandferb.fandom.com/wiki/Organization_Without_a_Cool_Acronym

The database tracks agents and their cover identities; the handlers and support personnel who manage them; the operations they participate in; the facilities and locations the operations actually happen; the gadgets checked out during missions; the nemeses being countered; and the “inators” they create. All these records help enable the sophisticated planning queries that OWCA operations demand.

2. Entity-Relationship Model



The ER diagram captures 15 entities and the relationships among them. Here it is summarized below:

Entity	Primary Key	Key Attributes
PERSONNEL	personnel_id	first_name, last_name, role_type, clearance_level
AGENT	agent_id	code_designation, species, date_recruited, is_alive, clearance_level
OPERATION	operation_id	op_codename, start_date, end_date, status, required_clearance
LOCATION	location_id	city, country
FACILITY	facility_id	facility_name, facility_type, required_clearance
LAIR	lair_id	lair_name, building_type, is_active
NEMESIS	nemesis_id	villain_name, evil_org, nemesis_level

INATOR	inator_id	inator_name, function_desc, weakness, current_status
GADGET	gadget_id	gadget_name, category
COVER_IDENTITY	cover_id	pet_name, host_family_name, is_burned
SKILL	skill_id	skill_name, skill_category
LANGUAGE	language_id	language_name
TRAINING_COURSE	course_id	course_name, duration_weeks, required_clearance
ASSIGNMENT	(operation_id, agent_id)	op_codename_in_op, role_in_op, assigned_date
GADGET_ASSIGNMENT	(gadget_id, agent_id, operation_id)	checked_out_date, returned_date

2.1 Key Relationships

- PERSONNEL Supervises OPERATION (one-to-many; the lead handler for each op)
- PERSONNEL Handles AGENT (many-to-many via HANDLES relationship; an agent can have multiple handlers over time)
- PERSONNEL Knows Identity of AGENT (many-to-many via KNOWS_IDENTITY; tracks who is read-in on an agent's true ID)
- AGENT is Assigned To OPERATION (many-to-many via ASSIGNMENT entity)
- AGENT has Cover Identity (one-to-one; each deployed agent has exactly one active cover)

- AGENT Checks Out GADGET for OPERATION (ternary via GADGET_ASSIGNMENT)
- AGENT Has Skill (many-to-many via AGENT_SKILL; includes proficiency level)
- AGENT Speaks Language (many-to-many via AGENT_LANGUAGE; includes fluency level)
- AGENT Trained In TRAINING_COURSE (many-to-many via TRAINED_IN; includes completion_date, grade)
- OPERATION Conducted At LOCATION (many-to-many via OPERATION_LOCATION; includes location_role)
- OPERATION Targets NEMESIS (many-to-one; each op has a primary nemesis)
- NEMESIS Operates From LAIR (one-to-one; each nemesis has a home lair)
- NEMESIS Invents INATOR (one-to-many; each inator belongs to one nemesis)
- FACILITY Located At LOCATION (many-to-one)
- LAIR Located At LOCATION (many-to-one)

3. Relational Model

Each entity and each many-to-many relationship maps to a table. The screenshots below denote that.

PERSONNEL

AGENT

OPERATION

LOCATION

FACILITY

LAIR

NEMESIS

INATOR

GADGET

COVER_IDENTITY

SKILL

LANGUAGE

TRAINING_COURSE

ASSIGNMENT (Agent-Operation)

GADGET_ASSIGNMENT (Gadget-Agent-Operation)

AGENT_SKILL

AGENT_LANGUAGE

TRAINED_IN

AGENT_LOCATION_HISTORY

Tracks where an agent lived over time (needed for queries like 'lived in London in 2021').

HANDLES

KNOWS_IDENTITY

OPERATION_LOCATION

4. Functional Dependencies

The following lists the non-trivial functional dependencies for each table.

Personnel

- personnel_id ---> first_name, last_name, role_type, clearance_level, current_location_id
- **AGENT**
 - agent_id ---> code_designation, species, date_recruited, is_alive, clearance_level, current_location_id, handler_id
 - code_designation ---> agent_id (code_designation is a candidate key)
- **OPERATION**
 - operation_id ---> op_codename, start_date, end_date, status, required_clearance, lead_handler_id, primary_nemesis_id
 - op_codename ---> operation_id (op_codename is a candidate key)
- **LOCATION**
 - location_id ---> city, country
- **FACILITY**
 - facility_id ---> facility_name, facility_type, required_clearance, location_id
- **LAIR**
 - lair_id ---> lair_name, building_type, is_active, location_id, nemesis_id
 - nemesis_id ---> lair_id (each nemesis has exactly one lair; candidate key)
- **NEMESIS**
 - nemesis_id ---> villain_name, evil_org, nemesis_level
- **INATOR**
 - inator_id ---> inator_name, function_desc, weakness, current_status, nemesis_id

- **GADGET**

- gadget_id ---> gadget_name, category

- **COVER_IDENTITY**

- cover_id ---> agent_id, pet_name, host_family_name, is_burned, location_id
- agent_id ---> cover_id (each agent has one cover; candidate key)

- **SKILL**

- skill_id ---> skill_name, skill_category
- skill_name ---> skill_id (candidate key)

- **LANGUAGE**

- language_id ---> language_name
- language_name ---> language_id (candidate key)

- **TRAINING_COURSE**

- course_id ---> course_name, duration_weeks, required_clearance
- ASSIGNMENT: (operation_id, agent_id) ---> op_codename_in_op, role_in_op, assigned_date
- GADGET_ASSIGNMENT: (gadget_id, agent_id, operation_id) ---> checked_out_date, returned_date
- AGENT_SKILL: (agent_id, skill_id) ---> proficiency
- AGENT_LANGUAGE: (agent_id, language_id) ---> fluency_level
- TRAINED_IN: (agent_id, course_id) ---> completion_date, grade
- AGENT_LOCATION_HISTORY: (agent_id, location_id, from_date) ---> to_date
- HANDLES: (personnel_id, agent_id) ---> start_date, end_date
- KNOWS_IDENTITY: (personnel_id, agent_id) ---> date_disclosed
- OPERATION_LOCATION: (operation_id, location_id) ---> location_role

5. Normalization (3NF / BCNF)

Every table in the OWCA schema satisfies at least Third Normal Form (3NF), and the majority satisfy Boyce-Codd Normal Form (BCNF). The argument proceeds as follows:

5.1 First Normal Form (1NF)

All tables have atomic, single-valued columns; no repeating groups. Skills, languages, and location history are represented in separate/

5.2 Second Normal Form (2NF)

2NF requires that every non-key attribute be fully dependent on the entire composite primary key (no partial dependencies). All composite-key tables in our schema satisfy this:

- ASSIGNMENT : op_codename_in_op and role_in_op depend on the full key (operation_id, agent_id), not on either alone.
- GADGET_ASSIGNMENT : both date columns depend on the full triple key.
- AGENT_SKILL : proficiency depends on (agent_id, skill_id) together.
- AGENT_LANGUAGE : fluency_level depends on the full pair.
- TRAINED_IN : completion_date and grade depend on (agent_id, course_id).
- AGENT_LOCATION_HISTORY : to_date depends on the full triple key.

5.3 Third Normal Form (3NF)

3NF requires no transitive dependencies

All tables satisfy this condition:

- **OPERATION:** lead_handler_id and primary_nemesis_id are FKs, not sources of transitive deps. Status, required_clearance, and date columns depend solely on operation_id.
- **AGENT:** handler_id is an FK reference; no non-key column determines another non-key column.
- **LAIR:** nemesis_id is both an FK and a candidate key: it determines lair attributes, but that is a BCNF dependency (see below), not a transitive dep through a non-key.

All other tables follow the same pattern: all non-key columns depend exclusively on the primary key, with no intra-row transitive paths.

5.4 BCNF Consideration

All tables are straightforwardly in BCNF!

7. Questions and Working Queries

8. Java GUI Description

The OWCA GUI is a NetBeans/Ant Java Swing application connecting to Oracle. It is structured as a JTabbedPane with three tabs, each simulating a different organizational role. Here, tab switching represents role switching.

8.1 Director Tab (DirectorPanel.java)

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Director Handler Analyst

Strategic filter

Status: **ALL** Min clearance: 1 Codename like: Refresh

Operation OpID Status Start End Clr Lead Handler Primary N... NLvl Roster

512	Tea Party	PLANNING	2025-02-20 ...		3	Stacy Hirano	Beavorius t...	2	1
510	Echo Bazaar	ACTIVE	2025-01-15 ...		4	Lyla Lolliberry	Lord Mortim...	5	2
509	Silent Sparrow	PLANNING	2024-09-01 ...		3	Norman Nor...	The Toucan...	3	2
507	Operation R...	ACTIVE	2024-04-12 ...		3	Gordon Grizzly	The Toucan...	3	1
504	X17	ACTIVE	2024-01-05 ...		5	Wanda Acro...	Lord Mortim...	5	3
505	Ghost Glacier	COMPROMI...	2023-11-20 ...	2024-02-28 ...	4	Stacy Hirano	Madame Py...	5	2
508	Crimson Cip...	COMPROMI...	2023-06-20 ...	2024-01-30 ...	4	Adyson Sive...	Otter the O...	4	2
503	Phoenix Pro...	CLOSED	2023-02-14 ...	2023-08-30 ...	4	Norman Nor...	Iguana Inqu...	4	2
511	ZZ8	CLOSED	2022-10-01 ...	2023-03-15 ...	5	Francis Mon...	Goldgorilla	5	2
506	Belladonna	CLOSED	2022-07-01 ...	2023-04-10 ...	4	Norman Nor...	Goldgorilla	5	2
501		ACTIVE	2022-03-01 ...		4	Lyla Lolliberry	Lord Mortim...	5	9

Roster Locations Gadgets

12 operations match

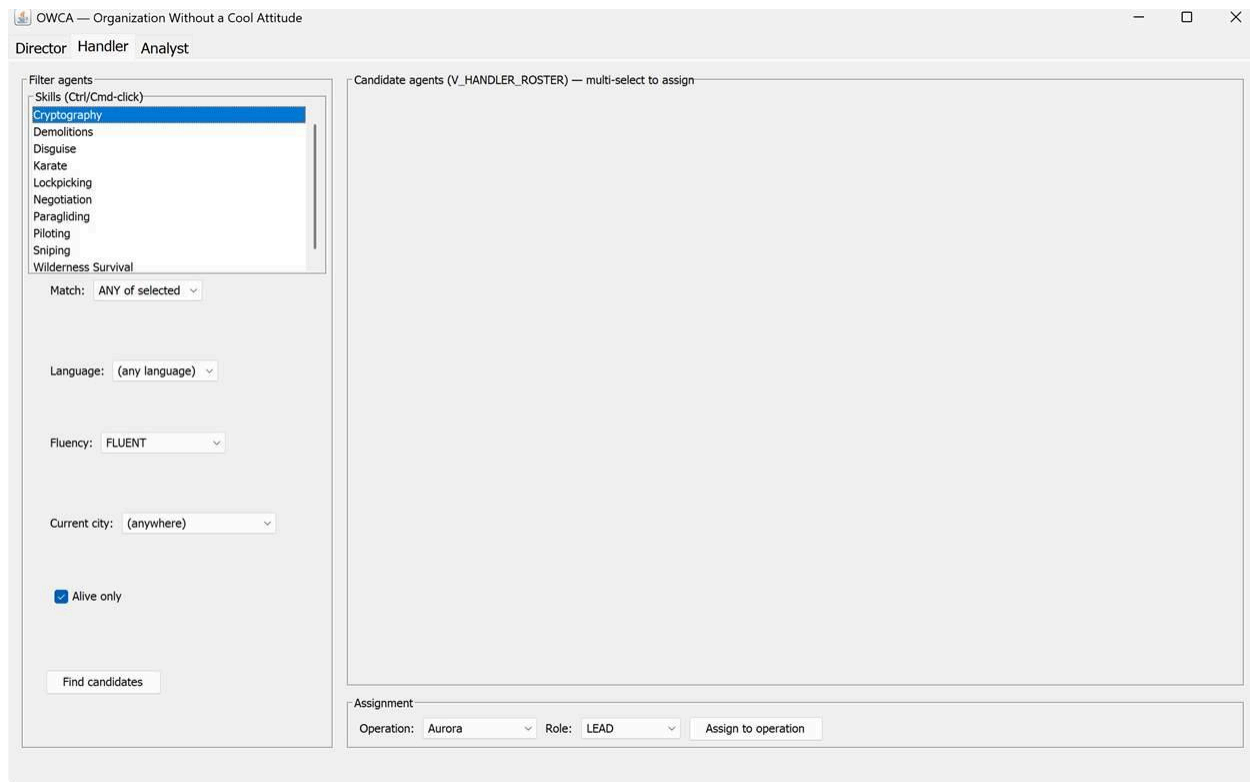
Role: Reviews all operations, assesses threats, checks rosters.

View used: V_DIRECTOR_OVERVIEW

Features:

- Filter strip: Status dropdown , minimum clearance spinner (1–5), and a codename search textbox. Any change to any filter re-runs the query.
- Operations master table: displays all matching operations with codename, status, dates, clearance, lead handler, nemesis name, nemesis threat level, and current roster size.
- Drill-down sub-tabs: Clicking any operation row auto-populates three sub-tabs: Roster (agents assigned, their roles and clearance), Locations (all sites used and their roles), and Gadgets (all equipment checked out for that op).

8.2 Handler Tab (HandlerPanel.java)



Role: Operational coordinator: Assembles mission teams and assigns agents to operations.

View used: V_HANDLER_ROSTER; writes to: ASSIGNMENT

Features:

- **Multi-select skill filter:** Match-mode toggle switches between ANY-of-selected and ALL-of-selected, enabling both broad and strict skill filtering.
 - **Language and fluency filter:** A dropdown picks a language; a second dropdown specifies minimum fluency.
 - **Location filter:** A dropdown restricts results to agents currently in a specific city.
- An 'Alive only' checkbox is checked by default.

- All filter components compose one dynamic SQL query. Then, results populate the candidates table showing skills and languages as aggregated text columns.
- **Assignment strip:** multi-select one or more candidate rows, pick an active/planning operation from the dropdown, pick a role, and click 'Assign to operation'. The panel uses a transactional batch INSERT with duplicate detection. The agents already on that operation are silently skipped and counted separately in the status bar.

8.3 Analyst Tab (AnalystPanel.java)

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Director Handler **Analyst**

Identity Trace Co-occurrence Damage Assessment Nemesis Profile

Choose agent

Code name: Agent E

Personnel who know the true identity

PersID	Name	Role	Disclosed
H00004	Wanda Acro...	HANDLER	2012-03-20 ...
H00001	Francis Mon...	DIRECTOR	2012-03-20 ...

Profile summary (V_ANALYST_PROFILE)

Species: Elephant | Alive: Y | Cover burned: N
 2 people know true identity | most recent op start: 2024-01-05
 Has been on a compromised op: NO | Paired nemesis: Dr. Eunice Earwig

Identity trace: Agent E

Role: Intelligence analyst: Handles deep-dive lookups on agents, operations, and threats.

View used: V_ANALYST_PROFILE (plus direct table queries)

The Analyst tab contains four sub-tabs:

- **Identity Trace:** Pick any agent from a dropdown; the panel shows every personnel member who knows that agent's true identity and when they were read in, plus a profile summary drawn from V_ANALYST_PROFILE.
- **Co-occurrence:** Pick two agents from dropdowns and click 'Find shared operations.
- **Damage Assessment:** dropdown populated with all COMPROMISED operations
- **Nemesis Profile:** Pick a nemesis from a dropdown; a split pane shows all inators that nemesis invented (with function, weakness, status) in the top pane and all operations run against that nemesis in the bottom pane.

9. Data Population

Tables were populated with AI-generated curation using Claude.

10. Submitted SQL Files

The following SQL scripts are included in the project submission and must be run in order on the S26_TEAM_2 Oracle account:

File	Description
01_create_tables.sql	Creates all 21 tables with constraints, primary keys, and foreign keys.
02_populate.sql	Inserts seed data into all tables.
03_queries.sql	Contains all 20 representative queries from Section 7

04_views.sql	Creates V_DIRECTOR_OVERVIEW, V_HANDLER_ROSTER, and V_ANALYST_PROFILE. Must be run before the GUI is launched.
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11. Running the Project

Prerequisites

- 20. Java 17 or higher
- 21. WIU VPN active The Oracle host is not reachable without a VPN. Or, you can just log in on WIU Secure.

Opening in NetBeans

- File ---> Open Project ---> select the OWCA_GUI folder ---> Open Project.
- NetBeans will auto-detect it as a Java with Ant project.
- Press the green Run arrow. The OWCA window opens with three tabs.

Connection Details

- User: S26_TEAM_2
- Password: bl68aQJc
- If the connection fails with 'Network Adapter' error, confirm VPN is active.
- If problems still remain, contact oor-yesufu@wiu.edu, j-roth@wiu.edu , r-ary@wiu.edu , and especially g-baramidze@wiu.edu .